J. Dairy Sci. 106:5054–5073
https://doi.org/10.3168/jds.2022-22965
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Colostrum insulin supplementation to neonatal Holstein bulls affects small intestinal histomorphology, mRNA expression, and enzymatic activity with minor influences on peripheral metabolism

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ABSTRACT

The objectives of this study were to evaluate how varying colostral insulin concentrations influenced small intestinal development and peripheral metabolism in neonatal Holstein bulls. Insulin was supplemented to approximately 5× (70.0 μg/L; n = 16) or 10× (149.7 μg/L; n = 16) the basal colostrum insulin (12.9 μg/L; BI, n = 16) concentration to maintain equivalent macronutrient intake (crude fat: 4.1 ± 0.06%; crude protein: 11.7 ± 0.05%; and lactose: 1.9 ± 0.01%) among treatments. Colostrum was fed at 2, 14, and 26 h postnatal and blood metabolites and insulin concentration were measured at 0, 30, 60, 90, 120, 180, 240, 360, 480, and 600 min postprandial respective to the first and second colostrum meal. At 30 h postnatal, a subset of calves (n = 8/treatment) were killed to excise the gastrointestinal and visceral tissues. Gastrointestinal and visceral gross morphology and dry matter and small intestinal histomorphology, gene expression, and carbohydrase activity were assessed. Insulin supplementation tended to linearly reduce the glucose clearance rate following the first meal, whereas after the second meal, supplementation linearly increased the rate of glucose absorption and nonesterified fatty acid clearance rate, decreased the time to maximum glucose concentrations, and decreased the time to reach minimum nonesterified fatty acid concentrations. Additionally, insulin clearance rate was linearly increased by insulin supplementation following the second colostrum feeding. However, there were no overall differences between treatments in the concentrations of glucose, nonesterified fatty acids, or insulin in plasma or serum. With respect to macroscopic intestinal development, dry rumen tissue mass linearly decreased when insulin was supplemented in colostrum, and supplementation linearly increased duodenal dry tissue density (g dry matter/cm) while tending to increase duodenal dry tissue weight. Increasing the colostrum insulin concentration improved small intestinal histomorphological development in the distal small intestine, as ileal villi height and mucosal-serosal surface area index were increased by supplementing insulin. Lactase enzymatic activity linearly increased in the proximal jejunum while ileal isomaltase activity linearly decreased with insulin supplementation. These data indicate that changes in colostrum insulin concentrations rapidly affect gastrointestinal growth prioritization and carbohydrase activity. The changes in gastrointestinal ontology result in minor changes in postprandial metabolite availability and clearance.
Key words: colostrum, insulin, neonate, bull, intestine

INTRODUCTION

Insulin is one of many bioactive factors present in greater concentrations in bovine colostrum than in milk (Fischer-Tlustos et al., 2021). Mean colostral insulin concentration is 50 to 100 times greater (Mann et al., 2016) than late gestation maternal peripheral blood concentrations, yet it exhibits high intersubject variability (6 to 327 μg/L; Malven et al., 1987; Aranda et al., 1991), possibly because late gestation factors such as parity (primi- vs. multiparous; Zinicola and Bicalho, 2019) and nutrient intake (Mann et al., 2016; Hare et al., 2022) can alter colostrum insulin concentrations. In addition, maternal-independent factors, such as time of milking relative to calving (i.e., colostrum vs. transition milkings; Zinicola and Bicalho, 2019) and heat-treatment (Mann et al., 2020), can affect the colostral insulin content. The maternal-dependent and maternal-independent variability in colostral insulin concentrations could be concerning, as the active transport of insulin against its concentration gradient by the mammary gland (Whitmore et al., 2012) supports the premise that colostral insulin content is important for neonatal calf development.

Received October 29, 2022.
Accepted December 29, 2022.
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We hypothesize that the gastrointestinal tract (**GIT**) could be a primary target of colostral insulin as the insulin receptor (**IR**) has been identified within the small intestine and colon (Pfaffl et al., 2002; Georgiev et al., 2003). Whereas IR abundance is greater in the proximal small intestine and colon, IR binding capacity is greatest in the ileum, particularly in response to multiple colostrum feedings (Georgiev et al., 2003). Insulin is thought to have a mitogenic effect on intestinal epithelial cells, as crypt cell proliferation increased in vitro (Arsenault and Ménard, 1984; Mah et al., 2014; Zhou et al., 2018) and is more promotive of proliferation than the effects of insulin-like growth factor-I (Zhou et al., 2018). Proteolysis in cultured cell lines has also been inhibited by insulin treatment (Ballard et al., 1982). Furthermore, insulin increases intestinal enzymatic activity in rodents (Ménard et al., 1981; Simon et al., 1982; Buts et al., 1990; Albert et al., 1994) and neonatal piglets (Shulman, 1990; Huo et al., 2006). Administering an anti-IR antibody to suckling rat pups inhibits insulin-stimulated mucosal development and sucrase and maltase activity (Marandi et al., 2001). Therefore, it is probable that variable colostrum insulin concentrations and ingestion can differentially affect intestinal development in neonatal calves. Considering that the immediate postnatal phase represents the shift from uteroplacental fetal nutrient delivery to nutrient ingestion and absorption by the GIT (Aynsley-Green, 1982; Girard et al., 1992), small intestinal growth and enzymatic capacity are critical to ensure the efficient digestion and absorption of nutrients to meet neonatal energy and protein requirements.

Neonatal plasma insulin concentrations increase rapidly in calves following the first colostrum meal and remain elevated during the first day postnatal respective to the following days (Oda et al., 1989; Hammon et al., 2000; Inabu et al., 2018). Some authors have postulated that this increase is in response to macronutrient consumption and improved GIT ontogenesis (Steinhoff-Wagner et al., 2011a, 2014). Alternatively, others suggest that the dramatic increase in plasma insulin concentration in response to colostrum consumption immediately after birth reflects absorption of whole, biologically active insulin while the GIT is in a state of enhanced permeability (Pierce et al., 1964; Kirovski et al., 2008). Other neonatal species have been shown to absorb insulin (Mosinger et al., 1959; Kelly, 1960; Asplund et al., 1962), but the capacity of the calf GIT to absorb colostral insulin is less certain (Grütter and Blum, 1991; Hammon et al., 2013). Authors that have demonstrated clear peripheral insulin increases (Kirovski et al., 2008) and hypoglycemic responses (Pierce et al., 1964; Kirovski et al.., 2008) in calves to oral insulin provision used pharmacological doses that exceeded the concentrations naturally occurring in colostrum. By contrast, others that did not observe the same hyperinsulinemic-hypoglycemic responses to oral insulin supplementation (Grütter and Blum, 1991) did not standardize colostral macronutrient intake between calves, thereby introducing a confounding factor respective to endogenous pancreatic insulin secretion (Nuttall et al., 1985). Even though it seems likely from prior studies that pharmacological concentrations of insulin are absorbed by the neonatal calf intestine, it is less clear if this would occur when insulin dose is similar to concentrations typically present in colostrum (~55 to 110 μg of insulin/L; Zinicola and Bicalho, 2019) and macronutrient intake is standardized between calves.

We first hypothesized that colostrum insulin would be intestinally absorbed, and that increasing colostrum insulin intake would induce hyperinsulinemia and hypoglycemia. Second, we hypothesized that increased colostral insulin ingestion would improve small intestinal histomorphological development and carbohydrase activity in neonatal Holstein bulls. Our objectives were to: (1) evaluate how peripheral glucose, nonesterified fatty acids (**NEFA**), and insulin concentrations differ with increasing colostrum insulin intake, and (2) evaluate how varying colostrum insulin concentrations affect small intestinal histomorphology, gene expression, and carbohydrase activity in neonatal Holstein bulls.

MATERIALS AND METHODS

The experiment was conducted in accordance with Canadian Council of Animal Care guidelines and approved by the University of Guelph (Guelph, ON) Animal Care Council (Animal Utilization Protocol #4126).

Neonatal Holstein bulls (n = 48; BW: 46.3 ± 0.8 kg) born between August 2019 and July 2020 (average temperature: 10.6°C; range: −10.5°C to 27.1°C; Weather Station ID: 4932, Hamilton A, ON) to multiparous Holstein-Friesian cows (median parity: 2; range: 1 to 6 prior pregnancies) on a commercial farm in Southwestern Ontario were used for this experiment. They remained with their dams for the first 30 to 60 min postnatal before enrollment by research personnel. During this time, they were carefully monitored to ensure that they did not consume colostrum directly from their dams. Once separated, they were weighed using a digital platform floor scale (PS2000, Brecknell Scale, Avery-Weigh Tronix), towel-dried, and transferred to calf pens (1.22 m^2) that were deeply bedded with wheat straw. Within the pens, all calves had visual, olfactory, and auditory stimulation from other cattle in their proximity.

The study was conducted as a completely randomized design, and research personnel randomized and

Table 1. Feeding rate, colostrum intake and composition, and characteristics of calves that consumed colostrum at 2, 14, and 26 h postnatal that had either basal insulin (BI; 12.9 μg/L) concentrations or was supplemented with an exogenous insulin to 5× (5BI; 70.0 μg/L) or 10× (10BI; 149.7 μg/L) the BI concentration

	Treatment				*P*-value	
Parameter	BI	5BI	10BI	SE	Linear	Quadratic
Feeding rate, %/kg BW	7.0	7.0	7.0	—	—	—
Meal mass, kg	3.2	3.3	3.2	0.06	0.78	0.58
Meal volume, L	3.1	3.1	3.1	0.06	0.66	0.59
Specific gravity, kg/L	1.04	1.04	1.04	0.002	0.36	0.52
Colostrum composition						
Crude fat, %	4.1	4.1	4.1	0.06	0.74	0.65
CP, %	11.7	11.7	11.6	0.05	0.74	0.61
Lactose, %	1.9	1.9	1.9	0.01	0.54	0.76
TS, %	19.6	19.6	19.6	0.08	0.81	0.81
Urea, mg/dL	35.4	35.1	35.2	0.35	0.64	0.69
BHB, m*M*	0.23	0.23	0.24	0.010	0.45	0.73
SCC, 1,000 cells/mL	148.2	138.0	136.1	7.30	0.21	0.70
IgG, g/L	63.9	63.5	64.2	1.19	0.86	0.76
Insulin,[1,2] μg/L	12.87	70.0	149.65	2.104	<0.001	<0.001
Calf characteristic						
Start BW, kg	45.9	46.6	46.3	0.84	0.70	0.64
Frame measurement, cm						
Body length	63.7	64.6	63.9	0.62	0.75	0.31
Chest circumference	84.2	85.2	84.5	0.58	0.65	0.24
Withers height	81.2	83.1	82.6	0.70	0.12	0.18
Hip height	84.0	86.4	86.1	0.84	0.054	0.24

[1]The endogenous colostrum insulin content for the 5BI and 10BI treatments is equivalent to the BI treatment.
[2]Colostrum insulin content was increased in the 5BI and 10BI treatments using 4 parts exogenous unlabeled bovine insulin (Sigma-I0516; Millipore-Sigma) and 1 part fluorescein isothiocyanate-labeled human insulin (Sigma-I3611; Millipore-Sigma) respective to an anticipated endogenous colostrum insulin content of 16.8 μg/L.

stratified treatments by birth order in advance of the study using the RAND function in Excel (Version 16.25, Microsoft Excel for Max, Office 365; Microsoft Canada, Microsoft Corporation). Before the experiment, it was determined that 16 calves per treatment were required to detect a 20% difference in metabolic characteristics with 80% statistical power at $P < 0.05$ (Berndtson, 1991). Over a 30-h period, nonblinded research personnel fed each calf 3 pasteurized colostrum meals that contained either: (1) basal colostrum insulin concentrations (**BI**, n = 16); (2) a 5-fold (**5BI**, n = 16) increase in colostrum insulin concentration respective to BI; or (3) a 10-fold (**10BI**, n = 16) increase in colostrum insulin concentration respective to BI. Blinding was not possible due to insulin was supplemented in colostrum at the time of feeding (described below). The composition of the basal colostrum was designed in collaboration with Saskatoon Colostrum Company Ltd. (Saskatoon, SK, Canada) by analyzing the insulin concentration of 189 colostrum samples with a commercial ELISA (Mercodia Bovine Insulin ELISA, Mercodia) and pooling and pasteurizing 112 colostrum sources that contained >40 g of IgG/L and low insulin concentrations (range: 1.7 to 62.5 μg of insulin/L; median: 18.6 μg of insulin/L). After pooling and pasteurization, the colostrum macronutrient, IgG, and insulin concentrations were re-analyzed and are reported in Table 1. The BI colostrum contained 16.8 μg of insulin/L and insulin concentration was supplemented in the 5BI and 10BI treatments using a 1:5 ratio of fluorescein isothiocyanate-labeled human insulin (Sigma-I3611; Millipore-Sigma) and unlabeled bovine insulin (Sigma-I0516; Millipore-Sigma). These doses (5BI and 10BI) were chosen to capture the range of insulin concentrations that have been observed in colostrum (Malven et al., 1987; Blum and Hammon, 2000; Zinicola and Bicalho, 2019). The fluorescein isothiocyanate-labeled human insulin was included as part of the supplemental insulin dose to distinguish exogenous colostrum insulin from endogenous pancreatic insulin by detection of the fluorescent label in plasma. The use of a pooled colostrum supplemented with insulin ensured that all calves consumed similar ratios of fat, CP, and lactose to reduce the probability of a confounding differential pancreatic insulin secretion among calves in response to differing macronutrient intake.

Colostrum was fed at 2.2, 14.1, and 26.1 h postnatal at a constant rate of 7% of BW across meals via an esophageal tube. We chose to use an esophageal tube for feeding to minimize the intercalf variation in the time per meal and reduce the confounding effect of rate of nutrient consumption on endogenous pancreatic

insulin secretion. Meal volume (3.10 ± 0.02 L) should have been sufficient to not hinder abomasal emptying and IgG absorption via tubing (Godden et al., 2009; Desjardins-Morrissette et al., 2018). To prepare meals, colostrum (4 L; stored frozen at −20°C) was thawed in a hot water bath for 1.5 to 2 h and warmed to 40°C for feeding. An excess volume of colostrum (109.0 ± 0.2% of what was needed for feeding) was prepared for each calf. Supplemental insulin was added and homogenized with the mixture by whisking for 1 min, then the precise mass of colostrum was given to each calf (7% of BW) and the remaining was used for determination of specific gravity. At each meal, a 40-mL colostrum sample was frozen at −20°C until analysis (described below).

Calves had jugular catheters placed aseptically by 1.2 h, as described in Fischer et al., 2018) to enable frequent blood collection throughout the duration of the experiment (30 h). Blood was removed via the jugular catheter at 0, 30, 60, 90, 120, 180, 240, 360, 480, and 600 min relative to the first and second colostrum feeding. The volume of blood collected from each calf per time point differed by calf BW (e.g., from 12 mL/time point for a 35 kg calf to 19 mL/time point for a 55 kg calf) but was standardized so that the total volume of blood collected over the 24 h sampling period was equivalent between calves respective to birth BW and did not exceed 1% of calf birth BW (vol/wt). Collection volume was standardized by calf BW for 2 reasons: (1) to ensure that there was adequate supernatant volume for all laboratory analyses, and (2) the calculation for apparent efficiency of absorption of IgG uses a constant plasma volume (9.1% of BW; Quigley et al., 1998), thus necessitating standardizing blood collection volume relative to BW to increase precision in calculating apparent efficiency of absorption between calves differing in BW (reported in Hare et al., 2023). After blood was collected, the catheter was flushed with saline and 2 mL of heparinized-saline (100 IU heparin/mL in 0.9% NaCl; Baxter JB1324, Baxter International) was infused to prevent blood from clotting within the catheter. Blood was then transferred to Vacutainers, one of which contained an anticoagulant (158 IU sodium heparin; BD366430, Becton Dickinson) for plasma separation, whereas the other (BD 366480, Becton Dickinson), intended for serum separation, did not. A competitive serine protease inhibitor (aprotinin; Sigma-A1153, Millipore-Sigma) was added to blood in heparin-coated Vacutainers at a rate of 5 μg/mL. Plasma Vacutainers were immediately centrifuged (920 × *g* for 25 min at 4°C), whereas serum Vacutainers were allowed to clot for 30 min at room temperature before centrifugation (920 × *g* for 25 min at 4°C). Once the plasma and serum had been separated, it was transferred in 3 aliquots (1.5 mL/aliquot) and stored in microcentrifuge tubes at −20°C until analysis. A fourth plasma aliquot (1 mL), intended for fluorescent intensity measurement, was stored in a black microcentrifuge tube at −20°C until analysis.

A subset of calves [n = 8/treatment; power based on required replication to detect differences in histomorphological measurements (Pyo et al., 2020)] randomly selected within treatment using the RANDOM function in Excel (Version 16.25, Microsoft Excel for Max, Office 365; Microsoft Canada, Microsoft Corporation) were stunned with a penetrative captive bolt gun and killed via pithing at 30 h 3 ± 8 min (mean ± SD) postnatal to excise the GIT. After exsanguination and confirmation of the absence of a heartbeat, the abdominal cavity was opened, and the GIT was ligated at the esophagus and rectum and removed along with the visceral organs. The GIT was landmarked according to Penner et al. (2014) and GIT segments (reticulorumen, omasum, abomasum, duodenum, proximal, mid, and distal jejunum, ileum, cecum, and colon) along with the spleen, kidneys (renal adipose removed), liver, and pancreas were weighed and subsampled (5 to 10 g) for DM analysis. The duodenum and ileum were respectively measured from the pyloric sphincter to the duodenal-colic fold and from the last branch of the cranial mesenteric artery to the ileo-cecal junction and the whole jejunum length was measured from the duodenal-colic fold to the last branch of the cranial mesenteric artery. From the duodenum and ileum, 30 cm lengths were partitioned to collect mucosal scrapings and tissue (5 cm) was preserved whole in 10% neutral-buffered formalin for histological analysis. Proximal and distal jejunal sections (30 cm in length for mucosal scrapings and 5 cm in length for histological preservation) were taken from the first and last 100 cm of the jejunum, respectively, and the mid jejunal section (30 cm in length for mucosal scrapings, 2 cm in length for histological preservation) was taken from the midpoint of the segment. Mucosal scrapings were collected as described by Burakowska et al. (2020).

Laboratory Analyses

Blood Metabolites and Insulin Concentration.

Plasma glucose (glucose oxidase/peroxidase reaction; Trinder, 1969) and serum NEFA (NEFA-HR (2) kit; Fujifilm Wako Chemicals) were measured in duplicate using enzymatic colorimetric analysis and values were accepted when the coefficient of variation (**CV**) was <5%. The inter- and intraassay CV were 1.4 and 1.6% (glucose) and 2.2 and 1.2% (NEFA), respectively. Plas-

ma insulin concentrations were quantified in duplicate with a commercial ELISA (Mercodia Bovine Insulin ELISA, Mercodia) with values accepted when the CV <7%. The inter- and intraassay CV were 6.6 and 2.5%, respectively. To measure fluorescein isothiocyanate-insulin fluorescence in plasma, the plasma was diluted at a factor of 1:2 plasma:phosphate buffered saline and emission was measured at 520 nm after excitation at 480 nm (Shen and Xu, 2000) using a fluorescent spectrophotometer.

Intestinal DM, Gene Expression, and Enzymatic Activity. Intestinal and visceral tissues (5 to 10 g) were dried at 105°C for ≥8 h until dry weight was consistent. Tissue DM (%) was calculated by expressing the tissue dry weight relative to the wet weight, multiplied by 100.

Snap-frozen mucosal scrapings from the duodenum, proximal jejunum, and ileum were finely ground using a mortar and pestle and kept super-cooled using liquid N_2 before and throughout grinding. Once the mucosa was ground to a homogeneous, fine powder, RNA was extracted using the PureLink RNA Mini Kit (Invitrogen, Thermo Fisher Scientific). Approximately 75 mg of super-cooled mucosal tissue was combined with 1 mL of TRIzol (Thermo Fisher Scientific) and homogenized using a vortex. Chloroform and 70% ethanol-alcohol were added to precipitate RNA and supernatants underwent successive spins (12,000 × *g*, 15 s, 22°C) and washes using a wash buffer and spin cartridge to bind, wash and elute RNA. Once RNA was bound, the spin cartridge membrane was dried by centrifugation (12,000 × *g*, 60 s, 22°C) and RNase-Free Water was added to the spin cartridge and incubated for 1 min. The RNA was eluted (12,000 × *g*, 120 s, 22°C) and its concentration was measured using UV-visible spectroscopy (Nanodrop One Microvolume, Thermo Fisher Scientific). Afterward, a DNase mixture [DNase I Reaction Buffer, RNase-free water, DNase I (amplification grade), and 1 mL of 25 m*M* EDTA; Thermo Fisher Scientific] was added to remove residual DNA during DNase digestion via heat inactivation. The RNA quality was assessed using its RNA integrity number (Schroeder et al., 2006) as measured by electrophoresis (TapeStation, Agilent Technologies). The average (±SE) RNA integrity number per region was 7.1 ± 0.11 (duodenum), 6.6 ± 0.10 (proximal jejunum), and 7.1 ± 0.10 (ileum). Next, cDNA was generated from 1 µg of RNA (High-Capacity cDNA Reverse Transcription Kit, Applied Biosystems) to analyze the gene expression of target genes relating to IR signaling (*INSR*, acc. no.: XM_002688832.6; *IRS1*, acc. no.: XM_003585773.5; *IRS2*, acc. no.: XM_025000170.1; *PI3K*, acc. no.: NM_174574.1; *Akt*, acc. no.: NM_173986.2; *S6K1*, acc. no.: NM_205816.1; and *mTOR*, acc. no.: XM_002694043.6), insulin degradation and trafficking (*IDE*, acc. no.: NM_001075849.1; *CEACAM1*, NM_205788.1), and glucose transport (*SLC5A1*, acc. no.: NM_174606.2; *SLC2A2*, acc. no.: NM_001103222) by quantitative real-time (**qRT**)-PCR. The house-keeping gene (*GAPDH*, acc. no.: NM_001034034.2) was determined to be stable and to not differ between treatments ($P > 0.05$; data not shown) and was selected as the endogenous control. The qRT-PCR was performed as described by Welboren et al. (2021a). Primer pair sequence, amplicon size, annealing efficiency, and coefficient of determination for target and endogenous house-keeping genes are reported in Table 2. The C_t value of target genes were normalized to the endogenous house-keeping geometric mean and relative gene expression was calculated as the fold change using the $2^{-\Delta\Delta Ct}$ method (Livak and Schmittgen, 2001; Schmittgen and Livak, 2008). Treatment means are expressed relative to the geometric mean of the BI treatment.

Small intestinal enzymatic activity was assessed as previously described (Trotta et al., 2020b,c). Mucosal scrapings from the duodenum, proximal jejunum, and ileum were weighed, diluted (2.0 mL of 9 g/L NaCl), and homogenized (Kinematica Polyton PT 10/35; Brinkmann Instruments Inc.) before measuring protein concentration (BCA procedure; Smith et al., 1985). Brush border carbohydrases (lactase, maltase, isomaltase, and glucoamylase) were assayed (Dahlqvist, 1964) using lactose, maltose, isomaltose, and soluble starch (as a substrate for glucoamylase; Kidder et al., 1972). Following incubation (described in Trotta et al., 2020a,b), the glucose end product was measured using the hexokinase/glucose-6-phosphate dehydrogenase procedure (Farrance, 1987) on a microplate spectrophotometer at 39°C. Data are expressed such that one unit (U) of enzymatic activity is equivalent to 1 (lactase and glucoamylase) or 0.5 µmol (maltase and isomaltase) glucose produced per minute of carbohydrase action and corrected for intestinal endogenous glucose (Trotta et al., 2020a,b). Enzyme activity is expressed as U/g tissue and U/g protein.

Brightfield Histological Measurements. Intestinal tissue (2 cm in length; duodenal, proximal and distal jejunal, and ileal tissue) was fixed in 40 mL of 10% neutral-buffered formalin (HT501128; Sigma-Aldrich) for 48 h and then preserved in 25 mL of 70% ethyl-alcohol until being sectioned onto glass slides and stained with hematoxylin and eosin (Animal Health Laboratory, Pathobiology Department, University of Guelph). At least 5 images were captured per slide at 40× magnification (Leica ICC50W; Leica Microsystems) and 15 intact villi and crypt attached to the submucosa were

Table 2. Primer sequences of the endogenous and target genes used for real-time PCR analysis in the duodenal, proximal jejunal, and ileal tissue of neonatal Holstein bulls fed colostrum at 2, 14, and 26 h postnatal that contained the basal insulin (16.8 μg/L) concentrations or was supplemented to increase insulin to 5 (83.8 μg/L) or 10 (167.5 μg/L) times that of the basal concentration

Gene	Gene name	Source[1]	Sequence,[2] 5′–3′	Amplicon size, bp	Efficiency,[3] %
GAPDH	Glyceraldehyde-3-phosphate dehydrogenase	NM_001034034.2	F: CCGTAACTTCTGGCC R: TTGATGGCGACGATGTCCACT	101	103
IR	Insulin receptor	XM_002688832.6	F: TGATGTGTACCCCTTGCCTG R: CTGGGCAGATGTCACCGAAT	89	105
IRS1	Insulin receptor substrate-1	XM_003585773.5	F: CCGGTGAGGACTTGAGCTAC R: AGGCGGTAGATGCCAATCAG	121	109
IRS2	Insulin receptor substrate-2	XM_025000170.1	F: ACGAGTACACCTTGATGCGG R: CCAATCTCGATGTCGCCGTA	127	94
PI3K	Phosphatidylinositol 3-kinase	NM_174574.1	F: CCGAAAGGGTGCTAAAGAGGA R: TGAGGTACTGGCCAAAGATTCAAAG	117	99
Akt	Protein kinase B	NM_173986.2	F: GCAGGACGTGTACGAGAAGAA R: TTGGTCAGGTGGCGTAATGG	124	106
S6K1	p70S6 kinase	NM_205816.1	F: TGACATGCTGACTGGAGCAC R: TGGCTTCTTGAGTGAGGTAG	110	103
mTOR	Mechanistic target of rapamycin kinase	XM_002694043.6	F: CTCCAACGACCCAGTTGTCA R: AGCAGTGAAAGTGTCCCCTG	79	108
IDE	Insulin degrading enzyme	NM_001075849.1	F: GGCTGCGTTAGGAATCATGC R: ACACAAACCATCCCCTGTCC	128	105
CEACAM1	CEA cell adhesion molecule 2	NM_205788.1	F: AACTCCAGCAGAAGCGACC R: TCACGATACCGACAATGGCA	100	108
SLGT1	Na^+-glucose cotransporter	NM_174606.2	F: CCATCTTCAACAGTGCCAGC R: ACCAACCGTCCTGCGATCAT	100	106
SLC2A2	Solute carrier family 2 member 2, aka GLUT2	NM_001103222	F: TCATGTCGGTGGGACTTGTG R: TGTCATGCTCACGTAATTCATCCAA	62	103

[1]National Center for Biotechnology Information (NCBI).

[2]F = forward; R = reverse.

[3]Efficiency = $-1 + 10^{(-1/\text{slope})} \times 100$.

measured (ImageJ 1.46r; National Institutes of Health) at random per calf (as described in Pyo et al., 2020). The mucosal surface area index was estimated from the mucosal-to-serosal amplification ratio (Kisielinski et al., 2002).

Calculations and Statistical Analyses

Postprandial Metabolite and Hormone Characteristics. Total and incremental area-under-the-curves (**AUC** and **I-AUC**; Chiou, 1978; Cardoso et al., 2011) were calculated for glucose and insulin over the 10-h measurement periods following colostrum feeding at 2.2 and 14.1 h postnatal. The decremental AUC (Schoenberg and Overton, 2011; Schoenberg et al., 2012) was calculated for postprandial NEFA concentrations during the same time intervals. To calculate the glucose availability rate, it was assumed that any increase in plasma glucose concentration following colostrum administration was due solely to the absorption of luminal glucose rather than hepatic gluconeogenesis. Given this assumption, glucose availability was calculated as the linear slope (using the slope function; Microsoft Excel for Mac, Version 16.64; Microsoft 365) from baseline (0 min) until maximal concentration ($\mathbf{C_{max}}$) was attained for each calf. Insulin appearance rate was similarly calculated but termed as "appearance" rate as we are unable to distinguish between intestinal absorption from pancreatic release and unable to differentiate the degree of hepatic insulin clearance from the portal vein. Glucose, NEFA, and insulin clearance rates were calculated as the linear slope (using the slope function; Microsoft Excel for Mac, Version 16.64; Microsoft 365) from C_{max} to nadir concentrations.

Statistical Models. As calves experienced unintended variation in environmental ambient temperature (described above), a discrete blocking factor was introduced post hoc to assess how ambient temperature influenced calf metabolic and hormonal responses. Rationale for assessing environmental ambient temperature was based on work by Stanko et al. (1991, 1992), wherein calves exposed to cold temperatures have greater plasma glucose following colostrum consumption (Stanko et al., 1992) and greater serum NEFA concentrations throughout cold exposure (Stanko et al., 1991). Block was further included for gastrointestinal-related variables to maintain consistency between statistical models. Calves were stratified into 5 discrete blocks based on the average ambient temperature (from hourly data over the 30 h from birth to experiment completion) to which they were exposed. Blocks ranged as such: −11.0 to −3.0°C, n = 4; −3.0 to 5.0°C, n = 15; 5.0 to 13.0°C, n = 5; 13.0 to 21.0°C, n = 12; and 21 to 29°C, n = 12; and calves from each treatment were represented within each bin.

All data were assessed as to whether their residuals met assumptions of having normal, independent, and identical distributions. Normality was assessed with PROC UNIVARIATE of SAS 9.4 (SAS Institute Inc.) and visual appraisal of boxplots. Data were confirmed normally distributed by a Shapiro-Wilk $P > 0.05$. When non-Gaussian residual distributions (Shapiro-Wilk $P < 0.05$) were detected, a lognormal distribution was specified in the MODEL statement of PROC GLIMMIX and the residuals were re-evaluated to determine if distribution normality had been established. Residual variance between treatments was tested for homogeneity using the COVTEST statement in PROC GLIMMIX and were confirmed homoscedastic when the residual homogeneity $P > \chi^2$ was >0.05. When the residual homogeneity $P > \chi^2$ was <0.05, heteroscedasticity was accounted for using the GROUP = command in the RANDOM statement of PROC GLIMMIX to specify that residual variance be grouped by treatment within the model.

Following assessment of residual distribution, data were analyzed as a completely randomized design with PROC GLIMMIX. The fixed effects of treatment (BI, 5BI, and 10BI) and block were included within the model and the random effect of calf (treatment) was included but specified to the R-sided random effects matrix. With respect to postprandial blood metabolite and insulin characteristics, baseline analyte concentrations preceding each postprandial period were tested as covariates within the model and included when $P < 0.05$; start BW was tested as covariate for gross intestinal and visceral morphology and included by the same condition. Unique contrast coefficients based on post hoc insulin supplementation generated using PROC IML were used to estimate the linear and quadratic treatment effects. In the case of repeated measurements, the fixed effects of time and the treatment by time interaction were included within the model. Covariance over time was modeled to determine the best-fit covariance structure (by the lowest Akaike information criteria and Bayesian information criteria values) per repeated variable. Degrees of freedom were approximated using Kenward-Rogers to account for unbalanced observations between treatments and Tukey-Kramer's post hoc adjustment was specified for nonrepeated measurements whereas Games-Howell's multiple comparison test was specified for unevenly-spaced repeated measurements (Lee and Lee, 2018) to identify means that differ. Differences were declared when $P < 0.05$ and tendencies were considered when $0.05 \leq P < 0.10$.

Table 3. Glucose, insulin, nonesterified fatty acids, and IgG curve characteristics for 10 h following the first colostrum feeding (2 h postnatal) when neonatal Holstein bulls consumed colostrum at 2, 14, and 26 h postnatal that had either basal insulin (BI; 12.9 μg/L) concentrations or was supplemented with an exogenous insulin to 5× (5BI; 70.0 μg/L) or 10× (10BI; 149.7 μg/L) the BI concentration

	Treatment				*P*-value	
Parameter[1]	BI	5BI	10BI	SE	Linear	Quadratic
Glucose						
Baseline, mg/dL	64.0	59.5	67.3	5.72	0.69	0.40
C_{max}, mg/dL	92.6	92.5	95.9	4.29	0.57	0.74
T_{max}, min	67.4	64.2	66.8	8.60	0.94	0.80
Absorption, mg/dL × min^{-1}	0.49	0.65	0.51	0.069	0.72	0.11
Clearance rate, mg/dL × min^{-1}	0.15	0.11	0.10	0.016	0.062	0.65
AUC, mg/dL × min	456.8	493.6	473.9	16.2	0.38	0.20
I-AUC, mg/dL × min	76.9	113.7	94.0	16.2	0.38	0.20
Insulin						
Baseline, μg/L	0.21	0.16	0.18	0.031	0.41	0.37
C_{max}, μg/L	2.19	2.56	2.38	0.279	0.56	0.46
T_{max}, min						
Appearance rate, μg/L × min^{-1}	0.023	0.014	0.018	0.0045	0.40	0.28
Clearance rate, μg/L × min^{-1}	0.0022	0.0039	0.0022	0.00051	0.92	0.014
AUC, μg/L × min	590.9	651.1	598.9	65.62	0.90	0.51
I-AUC, μg/L × min	430.7	432.6	457.1	62.83	0.76	0.88
Nonesterified fatty acids						
Baseline, mEq/L	1,033.3	953.8	1,014.5	93.15	0.85	0.58
C_{min}, mEq/L	205.0	191.3	220.1	17.88	0.55	0.35
$T_{min,}$ min	341.0	300.3	285.6	28.32	0.13	0.77
Clearance rate, mEq/L × min^{-1}	2.79	2.84	2.95	0.267	0.64	0.92
Decremental AUC, Eq/L × min	−338.5	−337.9	−341.9	12.80	0.84	0.88

[1]C_{max} = maximum concentration; T_{max} = time to maximum concentration; AUC = total area-under-the-curves; I-AUC = incremental area-under-the-curves.

RESULTS

Colostrum Intake and Calf Characteristics

Meal mass (3.2 ± 0.06 kg) and volume (3.1 ± 0.06 L) did not differ ($P \geq 0.58$; Table 1) among treatments. Colostral composition and specific gravity did not differ ($P \geq 0.21$), apart from colostral insulin concentration, which was greatest ($P < 0.001$) for 10BI (149.7 ± 2.10 μg/L) relative to 5BI (70.0 ± 2.10 μg/L) and BI (12.9 ± 2.10 μg/L) and greater for 5BI relative to BI. Treatment did not influence ($P \geq 0.12$) starting BW, body length, chest circumference, or withers height; however, hip height tended to be greater ($P = 0.054$) for 5BI and 10BI calves relative to BI calves.

Blood Metabolites and Plasma Insulin

Treatment quadratically affected ($P = 0.014$; Table 3) the insulin clearance rate during the first postprandial period when 5BI calves cleared insulin quicker than BI and 10BI calves. Glucose clearance rate tended to linearly decrease ($P = 0.062$) from the BI to 10BI treatment. Otherwise, plasma glucose and insulin and serum NEFA curve characteristics during the first postprandial period did not differ ($P \geq 0.11$) due to insulin treatment.

In the second postprandial period, the time to maximum plasma glucose concentration ($\mathbf{T_{max}}$) was reduced ($P = 0.024$; Table 4) by approximately an hour when 10BI was fed relative to BI. In contrast, glucose was absorbed at a greater ($P = 0.008$) rate for 10BI relative to BI. Plasma insulin C_{max} and clearance rate linearly increased ($P \leq 0.049$) as more insulin was fed in colostrum. The time to minimum ($\mathbf{T_{min}}$) serum NEFA linearly decreased ($P = 0.015$) with greater insulin supplementation as the NEFA clearance rate was linearly increased ($P = 0.013$).

Treatment and the treatment by time interaction did not influence ($P \geq 0.24$; Figure 1a-c) plasma glucose, serum NEFA, or plasma insulin concentrations throughout the first or second postprandial period. During the first postprandial period, plasma glucose concentrations increased (time: $P < 0.001$; Figure 1a) rapidly and peaked at 1 h postprandial, declined nadir at 6 h postprandial before a secondary postprandial rise at 8 and 10 h postprandial. By contrast, plasma glucose concentrations following the second colostrum feeding peaked (time: $P < 0.001$) from 2 to 4 h and decreased gradually until 10 h postprandial. Serum NEFA concentrations during the first postprandial period declined (time: $P < 0.001$; Figure 1b) rapidly from baseline until 4 to 6 h postprandial, then rose and equilibrated between 8 and 10 h postprandial. In

Table 4. Glucose, insulin, nonesterified fatty acids, and IgG curve characteristics for 10 h following the second colostrum feeding (14 h postnatal) when neonatal Holstein bulls consumed colostrum at 2, 14, and 26 h postnatal that had either basal insulin (BI; 12.9 μg/L) concentrations or was supplemented with an exogenous insulin to 5 × (5BI; 70.0 μg/L) or 10 × (10BI; 149.7 μg/L) the BI concentration

	Treatment				*P*-value	
Parameter[1]	BI	5BI	10BI	SE	Linear	Quadratic
Glucose						
Baseline, mg/dL	95.4	93.0	95.6	4.38	0.99	0.65
C_{max}, mg/dL	137.3	135.8	135.0	4.62	0.70	0.96
T_{max}, min	214.8	184.7	157.3	18.78	0.024	0.90
Absorption, mg/dL × min^{-1}	0.15	0.21	0.25	0.026	0.008	0.82
Clearance rate, mg/dL × min^{-1}	0.08	0.09	0.06	0.012	0.24	0.09
AUC, mg/dL × min	724.5	720.6	719.6	17.61	0.83	0.96
I-AUC, mg/dL × min	140.9	142.1	137.4	15.20	0.87	0.87
Insulin						
Baseline, μg/L	0.89	0.89	0.91	0.113	0.88	0.95
C_{max}, μg/L	1.60	2.23	2.00	0.174	0.049	0.064
T_{max}, min	97.5	108.2	113.6	14.17	0.39	0.90
Appearance rate, μg/L × min^{-1}	0.009	0.014	0.011	0.0020	0.32	0.11
Clearance rate, μg/L × min^{-1}	0.0021	0.0033	0.0031	0.00036	0.040	0.16
AUC, μg/L × min	601.5	741.0	691.1	56.33	0.21	0.20
I-AUC, μg/L × min	143.7	259.2	195.7	42.48	0.32	0.11
Nonesterified fatty acids						
Baseline, mEq/L	417.5	415.8	379.2	32.55	0.38	0.65
C_{min}, mEq/L	218.2	211.4	214.4	14.25	0.84	0.79
$T_{min,}$ min	238.1	256.4	163.4	23.93	0.015	0.051
Clearance rate, mEq/L × min^{-1}	0.92	0.98	1.43	0.147	0.013	0.25
Decremental AUC, Eq/L × min	−41.5	−47.6	−43.7	8.06	0.81	0.63

[1]C_{max} = maximum concentration; T_{max} = time to maximum concentration; AUC = total area-under-the-curves; I-AUC = incremental area-under-the-curves.

the second postprandial period, serum NEFA concentrations decreased (time: $P < 0.001$) from 0.5 to 3 h postprandial and gradually increased from 4 to 10 h postprandial. In period one, plasma insulin concentrations rose (time: $P < 0.001$; Figure 1c) from 0 to 1.5 h postprandial then decreased until 3 h postprandial. A secondary peak in plasma insulin during the first postprandial period was observed at 4 h postprandial, after which plasma insulin concentrations declined until reaching nadir at 8 h postprandial, increasing slightly by 10 h postprandial. Comparatively, plasma insulin peaked (time: $P < 0.001$) at 1.5 h postprandial during the second period, but in contrast declined gradually until 10 h, with no secondary peak observed.

Plasma fluorescence increased (time: $P < 0.001$; Supplemental Figure S1; https://doi.org/10.5683/SP3/XED4N5; Hare, 2023) after the first and second colostrum feeding but did not differ ($P \geq 0.85$) due to treatment or the treatment by time interaction.

Intestinal and Visceral Gross Morphology and Digesta Characteristics

Rumen dry tissue weight linearly decreased ($P = 0.010$; Table 5) for 10BI relative to BI calves (56.4 vs. 73.2 ± 4.00 g DM, respectively). Dry pancreas weight tended to be quadratically affected ($P = 0.063$) when calves consumed 5BI compared with BI. The dry duodenal weight tended to linearly increase ($P = 0.080$) with greater colostral insulin consumption, driven by a linear treatment effect ($P = 0.051$; Supplemental Table S1, https://doi.org/10.5683/SP3/XED4N5; Hare, 2023) of wet duodenal tissue weight rather than duodenal tissue DM% (22.7 to 23.4 ± 0.74% DM; $P = 0.58$; Supplemental Table S1). Consequently, intestinal tissue density increased linearly ($P = 0.042$; Supplemental Table S1) with supplemental colostrum insulin concentration as duodenal length did not differ ($P = 0.45$) between treatments. The ileum and cecum were linearly shorter ($P \leq 0.044$) when calves consumed more insulin in colostrum. Otherwise, the remaining dry tissue weights and jejunal and colonic lengths were not affected ($P \geq 0.14$) by colostral insulin supplementation.

Digesta osmolality and pH were not affected by treatment ($P \geq 0.13$; Supplemental Table S2; https://doi.org/10.5683/SP3/XED4N5; Hare, 2023), apart from mid jejunal pH (quadratic: $P = 0.008$).

Small Intestinal Histomorphometry

Duodenal villi were longest ($P = 0.041$; Table 6) when calves received 10BI rather than 5BI. Ileal villi height was greatest ($P = 0.001$) for 10BI versus BI calves by 215.4 ± 38.28 μm and, similarly, distal jejunal villi height tended to linearly increase ($P = 0.091$)

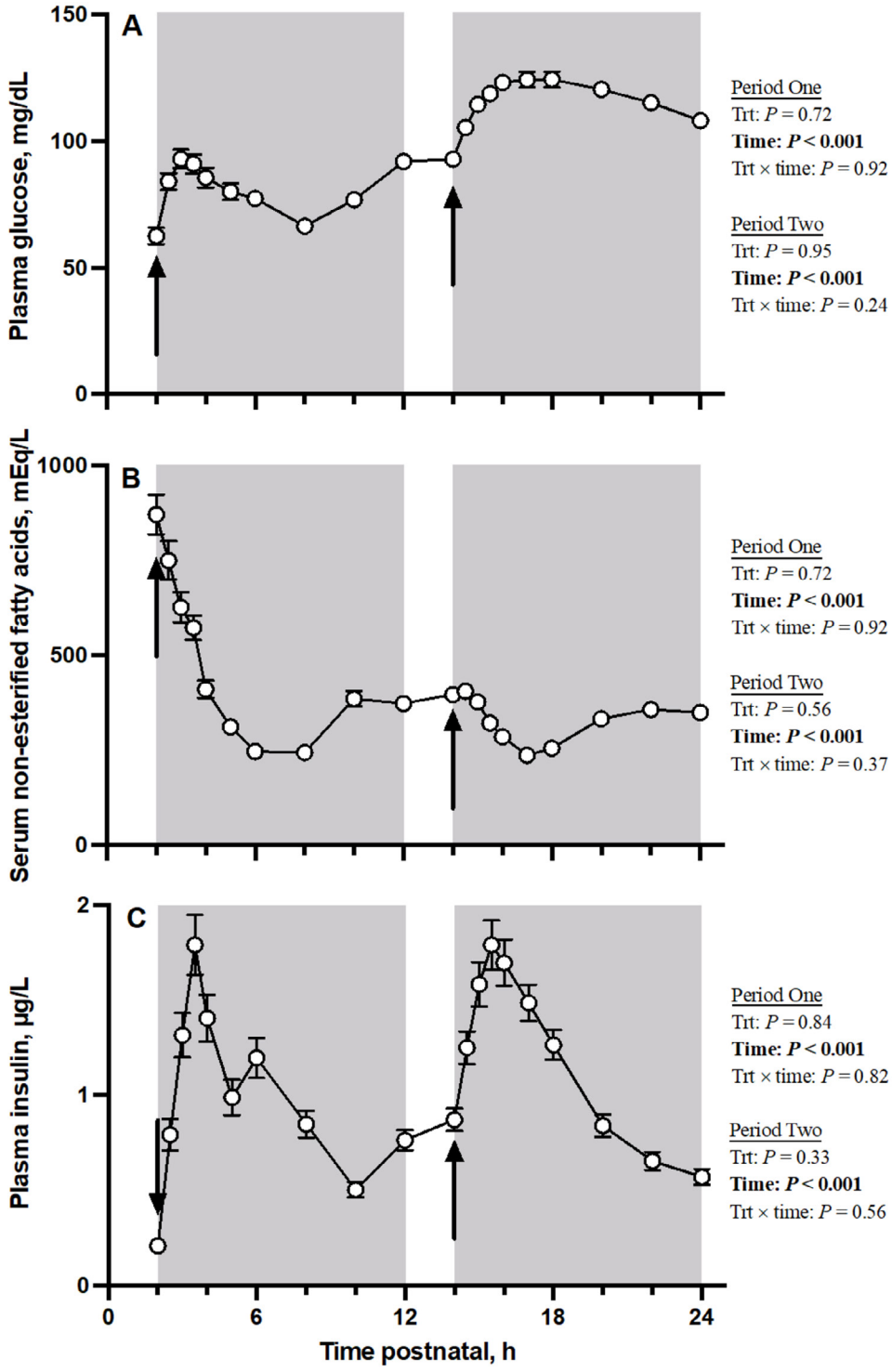


Figure 1. Plasma glucose (panel A; SE: 2.94 mg/dL), serum nonesterified fatty acids (panel B; SE: 20.97 mEq/L), and plasma insulin concentrations (panel C; SE: 0.09 µg/L) of neonatal Holstein bulls that consumed colostrum at 2, 14, and 26 h postnatal that had either basal insulin (BI; 12.9 µg/L) concentrations or was supplemented with an exogenous insulin to 5× (5BI; 70.0 µg/L) or 10× (10BI; 149.7 µg/L) the BI concentration. Period 1 is the postprandial period following the first meal at 2 h postnatal, whereas period 2 is the postprandial period following the second meal at 14 h postnatal. Data are presented as the effect of time; that is, the average of BI, 5BI, and 10BI analyte concentrations at individual time points. Black arrows denote when meals were fed at 2 and 14 h postnatal. The concentrations of glucose, nonesterified fatty acids, insulin, and IgG all differed with respect to time ($P < 0.001$) during the first and second postprandial period but were not different ($P \geq 0.24$) due to the effect of treatment or the treatment by time interaction. Data are presented as mean ± SE and some error bars are too small to be seen.

Table 5. Intestinal and visceral gross morphology of neonatal Holstein bulls that consumed colostrum at 2, 14, and 26 h postnatal that had either basal insulin (BI; 12.9 μg/L) concentrations or was supplemented with an exogenous insulin to 5 × (5BI; 70.0 μg/L) or 10 × (10BI; 149.7 μg/L) the BI concentration

	Treatment				*P*-value	
Parameter	BI	5BI	10BI	SEM	Linear	Quadratic
Tissue weight, g of DM						
Rumen	73.2	64.7	56.4	4.00	0.010	0.90
Omasum	2.9	2.9	2.9	0.12	0.65	0.98
Abomasum	63.2	61.8	67.2	3.58	0.48	0.42
Duodenum	10.1	10.7	12.7	0.98	0.080	0.54
Jejunum	255.2	238.5	249.1	13.34	0.72	0.42
Ileum[1]	33.7	38.0	34.2	3.26	0.83	0.14
Cecum	10.8	10.2	9.9	1.13	0.58	0.94
Colon	75.6	75.3	71.0	7.55	0.69	0.81
Liver	267.4	261.6	255.5	8.82	0.38	0.96
Spleen[1]	25.8	27.2	25.4	1.90	0.89	0.29
Pancreas[1]	8.7	11.7	10.7	1.22	0.12	0.063
Kidneys[1]	45.8	46.7	48.0	2.90	0.46	0.89
Intestinal length, cm						
Duodenum	72.6	72.7	75.6	2.63	0.45	0.64
Jejunum	1,673.5	1,646.1	1,617.5	48.93	0.44	0.96
Ileum	141.6	141.9	122.5	6.00	0.044	0.16
Cecum	31.7	24.9	23.6	2.58	0.040	0.44
Colon	196.3	201.6	204.5	7.53	0.46	0.92

[1]BW was included in the model as a covariate. The *P*-values for BW as a covariate are as follows: ileum, $P = 0.047$; spleen, $P = 0.021$; pancreas, $P = 0.024$; and kidneys, $P = 0.001$.

when colostrum insulin concentration was increased. In the distal jejunum, the linear response ($P \leq 0.038$) to treatment was present with respect to villi width, crypt depth, and crypt width that all linearly increased with increasing colostral insulin content. The surface area index was greatest ($P = 0.002$) in the ileum for calves

Table 6. Histomorphometric measurements in the duodenum, proximal and distal jejunum, and ileum of neonatal Holstein bulls that consumed colostrum at 2, 14, and 26 h postnatal that had either basal insulin (BI; 12.9 μg/L) concentrations or was supplemented with an exogenous insulin to 5× (5BI; 70.0 μg/L) or 10× (10BI; 149.7 μg/L) the BI concentration

	Treatment				*P*-value	
Parameter	BI	5BI	10BI	SE	Linear	Quadratic
Villi height, μm						
Duodenum	547.1	529.5	756.7	63.43	0.041	0.097
Proximal jejunum	622.7	638.2	699.2	46.96	0.29	0.65
Distal jejunum	721.8	729.1	847.4	46.95	0.091	0.29
Ileum	675.1	772.6	890.5	38.28	0.001	0.69
Villi width, μm						
Duodenum	93.5	94.0	112.9	7.66	0.11	0.29
Proximal jejunum	119.9	123.9	128.6	5.84	0.33	0.92
Distal jejunum	118.6	126.8	133.6	4.27	0.027	0.98
Ileum	131.7	142.4	140.6	4.79	0.30	0.31
Crypt depth, μm						
Duodenum	191.3	182.5	229.4	23.67	0.31	0.31
Proximal jejunum	173.9	189.6	192.0	7.67	0.12	0.52
Distal jejunum	177.8	193.6	208.5	8.67	0.026	0.94
Ileum	180.0	194.9	204.9	11.62	0.15	0.92
Crypt width, μm						
Duodenum	40.4	42.1	47.0	3.12	0.12	0.68
Proximal jejunum	53.1	55.0	52.7	1.51	0.90	0.35
Distal jejunum	46.3	49.5	51.1	1.48	0.038	0.73
Ileum	48.8	51.1	52.1	1.70	0.19	0.78
Surface area						
Duodenum	13.7	13.2	16.1	1.06	0.17	0.17
Proximal jejunum	12.2	12.1	13.2	0.65	0.34	0.47
Distal jejunum	15.1	14.3	15.8	0.65	0.51	0.16
Ileum	13.0	14.0	16.1	0.60	0.002	0.35

Table 7. mRNA expression of genes involved in the insulin signal transduction pathway, intracellular insulin degradation, and glucose transport in the duodenum, proximal jejunum, and ileum of neonatal Holstein bulls that consumed colostrum at 2, 14, and 26 h postnatal that had either basal insulin (BI; 12.9 μg/L) concentrations or was supplemented with an exogenous insulin to 5× (5BI; 70.0 μg/L) or 10× (10BI; 149.7 μg/L) the BI concentration

	Treatment				*P*-value	
Parameter	BI	5BI	10BI	SE	Linear	Quadratic
IR						
Duodenum	1.06	0.64	0.71	0.075	0.027	0.014
Proximal jejunum	1.00	1.08	1.00	0.070	0.95	0.36
Ileum	1.03	1.17	0.80	0.237	0.51	0.42
IRS1						
Duodenum	1.02	0.71	0.66	0.097	0.024	0.38
Proximal jejunum	1.03	0.67	0.65	0.086	0.008	0.14
Ileum	1.03	1.18	0.82	0.174	0.47	0.22
IRS2						
Duodenum	1.06	0.99	0.66	0.087	0.041	0.15
Proximal jejunum	1.08	1.38	1.22	0.186	0.61	0.32
Ileum	1.02	0.92	0.91	0.156	0.62	0.83
PI3K						
Duodenum	1.04	0.79	0.79	0.089	0.062	0.27
Proximal jejunum	1.01	0.91	1.06	0.106	0.78	0.31
Ileum	1.01	1.38	1.23	0.341	0.60	0.55
Akt						
Duodenum	1.01	0.70	0.72	0.062	0.006	0.066
Proximal jejunum	1.00	0.92	1.22	0.120	0.27	0.18
Ileum	0.60	0.67	0.57	0.057	0.77	0.21
S6K1						
Duodenum	1.02	0.74	0.84	0.068	0.046	0.098
Proximal jejunum	1.03	1.08	1.48	0.183	0.13	0.39
Ileum	1.01	1.33	1.42	0.255	0.20	0.73
mTOR						
Duodenum	1.05	0.84	0.74	0.770	0.011	0.64
Proximal jejunum	0.77	0.75	0.79	0.071	0.84	0.72
Ileum	1.13	2.20	1.31	0.341	0.52	0.09
IDE						
Duodenum	1.02	0.76	0.85	0.072	0.11	0.069
Proximal jejunum	1.01	0.88	0.99	0.073	0.84	0.20
Ileum	1.00	1.06	1.12	0.141	0.60	0.99
CEACAM1						
Duodenum	1.02	0.88	0.95	0.092	0.57	0.36
Proximal jejunum	1.18	0.79	1.24	0.228	0.94	0.10
Ileum	0.96	1.27	1.09	0.220	0.64	0.38
SGLT1						
Duodenum	1.01	0.69	1.01	0.089	0.90	0.007
Proximal jejunum	1.05	0.88	1.39	0.194	0.28	0.14
Ileum	1.00	1.41	1.12	0.306	0.74	0.36
SLC2A2						
Duodenum	1.11	0.93	0.97	0.146	0.50	0.53
Proximal jejunum	1.15	1.16	1.54	0.214	0.25	0.43
Ileum	1.06	1.14	1.11	0.534	0.95	0.94

that consumed 10BI rather than BI. No differences ($P \geq 0.11$) were observed respective to proximal jejunal villi height, villi width, crypt depth, and crypt width or surface area index.

Gene Expression

In the duodenum, the mRNA expression of the IR decreased linearly ($P = 0.027$; Table 7) with insulin supplementation. Likewise, duodenal and proximal jejunal mRNA expression of *IRS1* decreased ($P \leq 0.024$) and duodenal *IRS2* mRNA expression decreased ($P = 0.041$) when colostrum was supplemented with insulin. A tendency for a linearly reduction ($P = 0.062$) in duodenal *PI3K* mRNA expression was observed from BI to 10BI, whereas *Akt*, *S6K1*, and *mTOR* mRNA expressions were linearly reduced ($P \leq 0.046$). A tendency for a quadratic effect ($P = 0.069$) of treatment on duodenal *IDE* mRNA expression was observed. Duodenal *SGLT1* mRNA expression was quadratically effected ($P = 0.007$) by treatment. Otherwise, mRNA expression of *CEACAM1* and *SLC2A2* was not influenced ($P \geq 0.10$) by insulin treatment in any small intestinal region.

Table 8. Enzyme activity and relative enzyme activity in the duodenum, proximal jejunum, and ileum of neonatal Holstein bulls that consumed colostrum at 2, 14, and 26 h postnatal that had either basal insulin (BI; 12.9 μg/L) concentrations or was supplemented with an exogenous insulin to 5× (5BI; 70.0 μg/L) or 10× (10BI; 149.7 μg/L) the BI concentration

	Treatment				*P*-value	
Parameter	BI	5BI	10BI	SE	Linear	Quadratic
Enzyme activity, U/g tissue						
Maltase						
Duodenum	0.18	0.21	0.20	0.043	0.63	0.74
Proximal jejunum	0.20	0.31	0.27	0.068	0.47	0.39
Ileum	0.54	0.44	0.54	0.085	0.93	0.33
Isomaltase						
Duodenum	1.13	1.12	1.07	0.035	0.28	0.56
Proximal jejunum	1.01	1.05	1.07	0.071	0.55	0.96
Ileum	1.13	1.12	0.97	0.049	0.043	0.19
Lactase						
Duodenum	118.45	119.79	115.47	2.931	0.52	0.41
Proximal jejunum	105.40	116.32	123.34	2.023	<0.001	0.61
Ileum	114.00	113.88	116.91	3.076	0.55	0.65
Glucoamylase						
Duodenum	0.77	0.93	0.63	0.113	0.46	0.096
Proximal jejunum	0.84	0.61	0.60	0.159	0.32	0.62
Ileum	0.65	0.85	0.89	0.222	0.45	0.81
Specific enzyme activity, U/g protein						
Maltase						
Duodenum	1.9	2.1	2.6	0.74	0.39	0.81
Proximal jejunum	2.0	3.0	2.8	0.68	0.43	0.44
Ileum	3.7	3.3	3.9	0.89	0.90	0.49
Isomaltase						
Duodenum	12.1	11.3	11.3	0.47	0.27	0.51
Proximal jejunum	10.1	10.2	10.4	0.89	0.83	0.98
Ileum	7.4	8.1	7.3	0.90	0.97	0.47
Lactase						
Duodenum	1,264.3	1,169.8	1,222.9	45.61	0.51	0.20
Proximal jejunum	1,054.8	1,137.3	1,192.2	56.11	0.24	0.90
Ileum	780.8	851.4	883.6	60.28	0.25	0.82
Glucoamylase						
Duodenum	8.1	9.3	6.7	1.18	0.46	0.18
Proximal jejunum	8.3	6.0	6.2	1.53	0.36	0.52
Ileum	2.8	5.7	4.4	1.52	0.33	0.30

Small Intestinal Carbohydrase Activity

Ileal isomaltase activity linearly decreased ($P = 0.043$; Table 8) for 10BI relative to BI calves. Proximal jejunal lactase was greatest ($P < 0.001$) for 10BI as compared with BI treatments, linearly increasing ($P < 0.001$) with insulin supplementation. Maltase and glucoamylase activities did not differ ($P \geq 0.096$) within any small intestinal region. When carbohydrase activity was standardized relative to total protein content, no treatment linear or quadratic differences ($P \geq 0.18$) were present in the relative carbohydrase activity in any intestinal region.

DISCUSSION

Although the bioactive properties of colostrum as a whole on the neonatal bovine GIT and metabolism have been investigated (Steinhoff-Wagner et al., 2011a,b, 2014, 2015; Pyo et al., 2020), we chose to evaluate how isolated changes in colostrum insulin concentration affected peripheral metabolism and GIT development in neonatal Holstein bulls. Our objective was to increase colostral insulin concentration to 5 and 10 times that of the basal concentration while maintaining the fat, CP, and lactose content equivalent between treatments. Supplementation thresholds (5 and 10 times the BI) were chosen to maintain colostrum insulin concentration within the observed biological range (Malven et al., 1987; Blum and Hammon, 2000; Zinicola and Bicalho, 2019) and provide a linear dose. We dosed exogenous insulin sources respective to an anticipated basal insulin concentration of 16.8 μg/L using a combination of bovine insulin and a fluorescent-labeled human insulin. Retrospective colostrum analysis indicated that basal insulin concentration was lower than anticipated (12.9 rather than 16.8 μg/L), resulting in our intended relative supplementation thresholds exceeding the 5BI and 10BI treatments by 5.4 and 11.7× that of the BI. Even though it is unclear as to why BI concentrations were

less than previously measured, the marginal decrease in basal insulin concentration is negligible and we maintain a linear increase in colostrum insulin concentration across our treatments. In addition, retrospective colostrum analysis demonstrates that BI, 5BI, and 10BI were provided colostrum containing similar macronutrient concentrations. Therefore, our experimental model of supplementing a frozen, pooled colostrum source with exogenous bovine insulin was appropriate to evaluate the objectives of this study.

We hypothesized that the neonatal GIT would be a primary target of colostral insulin (Pfaffl et al., 2002; Georgiev et al., 2003; Georgieva et al., 2003), and that IR binding and stimulation would lead to villi growth and crypt development. The linear increase in duodenal and ileal villi length and ileal surface area index increasing linearly in response to insulin supplementation in colostrum supports our hypothesis that colostral insulin affects the neonatal bovine GIT and alters histomorphological development. Improvements in intestinal growth and enterocyte proliferation in vitro (Arsenault and Ménard, 1984; Marandi et al., 2001; Zhou et al., 2018) and in vivo (Shulman, 1990; Shamir et al., 2005) have been previously documented. Further, the tendencies and linear responses observed in the distal jejunum with respect to villus height and width and crypt depth and width additionally support our hypothesis.

Interestingly, prior studies documented improved villus development (murine model; Shamir et al., 2005) and crypt cell proliferation (primate model; Wheeler and Challacombe, 1997) predominantly in the duodenum. Comparatively, our responses in histomorphological development were observed primarily in the distal intestine. Our observed results in small intestinal histomorphometry align with previously reported fluctuations in binding capacity (Georgiev et al., 2003) and high-affinity binding sites (Hammon and Blum, 2002) for insulin along the neonatal bovine GIT.

Given the short study interval (30 h postnatal) in the present study and that calves only received 3 feedings of colostrum, we did not expect that GIT gross morphometry differences would be observed. It is unclear why colostral insulin supplementation linearly decreased dry ruminal tissue mass, particularly because the rumen contributes a relatively small portion of the GIT and is relatively underdeveloped in preweaning ruminants (Meale et al., 2017). We speculate that colostral insulin concentration altered nutrient partitioning and growth was prioritized between different GIT segments. We propose this as the multiple SI responses with insulin supplementation (i.e., histomorphometric improvements within the distal small intestine, the linear increase in duodenal villi height, and the linear increase in dry intestinal density) are negatively reciprocal to the linear decrease in dry ruminal tissue mass. Although ileal and cecal lengths decreased with insulin supplementation, dry ileal and cecal weights and density were unaffected, demonstrating that changes in intestinal length were unsubstantial. Because all calves had similar gross energy and nutrient intake, it seems that colostral insulin promoted the utilization of nutrients within the small intestine rather than the foregut. Intestinal mass was observed to increase in insulin-supplemented neonatal piglets without changes in length or mass of other intestinal regions (Shulman, 1990; Huo et al., 2006). Even though it is known that differences in macronutrient source and digestibility can alter gut morphology in neonatal and preweaning calves (Montagne et al., 1999; Welboren et al., 2021a), we are unaware of other studies demonstrating that colostrum bioactive compounds can alter intestinal nutrient partitioning when gross macronutrient and energy consumption is consistent between calves. This could be explained, in part, by the linear increase in proximal jejunal lactase activity with increasing colostral insulin concentration plausibly increasing the digestible energy supply from colostrum.

To our knowledge, we are the first to demonstrate that carbohydrase activity in the neonatal calf intestine is responsive to insulin supplementation by linearly decreasing ileal isomaltase activity and linearly increasing proximal jejunal lactase activity – particularly, because our experimental model was not confounded by inequivalent nutrient intake (Shulman and Burrin, 1991). Lactase activity increased in the proximal jejunum, but not the duodenum or ileum, as corroborated by prior studies demonstrating that lactase was most active in the proximal jejunum (Steinhoff-Wagner et al., 2015; Trotta et al., 2020a). Although the lack of response in maltase or glucoamylase activity to insulin supplementation in this study contrasts previous data in suckling rodents (Arsenault and Ménard. 1984; Buts et al., 1990, 1998; Marandi et al., 2001), it highlights that intestinal enzymatic response to insulin treatment varies between species (Ménard et al., 1981; Simon et al., 1982; Shulman, 2002). More importantly, it emphasizes the importance of rapid induction of lactase activity in the neonatal calf.

Because insulin has been confirmed to transduce its signal through the IR (Marandi et al., 2001) and intraluminal insulin binds to its apical receptor on villi before transcytosis (Reis et al., 2008), we endeavored to analyze whether colostral insulin treatment affected the mRNA expression of genes involved in the insulin signaling pathway in the GIT. We found that multiple genes within the insulin signaling pathway (*IR*, *IRS1*, *IRS2*, *PI3K*, *Akt*, *S6K1*, and *mTOR*) were affected by colostral insulin concentration, predominantly within the duodenum, such that increasing BI to 10BI sup-

pressed mRNA expression. Speculatively, the lack of response in proximal and ileal mRNA expression may be due to insulin digestion in the proximal small intestine that would suggest that histomorphological growth stimulation should occur in advance of maturation in proteolytic activity. Yet, it seems counter-intuitive that greater colostral insulin intake would suppress mRNA expression of the *IR* and downstream genes given the histomorphological development and lactase responses to 10BI supplementation. However, this response appears to be consistent with oral insulin provision (*IR* mRNA downregulation in intestinal tissue; Huo et al., 2006) and hyperinsulinemia in vitro (*IR* signaling pathway protein abundance; Cen et al., 2022). Ligand-dependent repression of mRNA transcription can occur via multiple mechanisms (Santos et al., 2011) and *FOXO1* and *SIN3A* are involved in *IR* pathway suppression (Cen et al., 2022). That said, we have not investigated the mRNA expression of these genes herein and are unable to confirm that they are involved in insulin-stimulated *IR* pathway downregulation. Furthermore, it is unclear how downregulation of the insulin signaling pathway influences the GIT of calves >30 h old, but, speculatively, it may indicate that multiple colostrum feedings could be detrimental to *IR* pathways in the GIT and that feeding transition milk (which has a lower insulin concentration; Blum and Hammon, 2000) might be preferred to attenuate this response.

As evidenced in this study and others, blood glucose concentrations are low in neonatal calves immediately after birth before colostrum intake and increase to normoglycemia by 24 h (Knowles et al., 2000; Kirovski et al., 2011; Renaud et al., 2022). Neonatal calves have limited glycogen reserves before colostrum intake that can support summit metabolism for ≤3 h (Okamoto et al., 1986) and relative underdeveloped gluconeogenic capacity that increases as calves age (Steinhoff-Wagner et al., 2011b). Colostrum lactose consumption is particularly important for neonatal calf glucose homeostasis and glucose availability, which in turn, are critical for immune response (Kvidera et al., 2017). Increases in intestinal lactase activity, as promoted by colostrum insulin supplementation, appears to facilitate the rate of lactose digestion and glucose absorption in neonatal calves, supporting glucose homeostasis and immune responsiveness in calves. Glucose metabolic parameters (T_{max} and absorption rate) during the second postprandial support that 5BI and 10BI calves digested lactose and absorbed glucose more rapidly than BI calves and that the rate of digestion was linear respective to insulin supplementation. Serum NEFA T_{min} and clearance rate following the second meal mirror the glucose metabolic response, further supporting that insulin is improved the glycemic and energetic state of neonatal calves. These responses occur during the second postprandial period rather than the first, indicating that the neonatal intestine had adapted to the insulin stimulation during the 12 h intermeal interval. Overall, total lactose digestion and glucose absorption is presumably equivalent among treatments because plasma glucose C_{max}, AUC, and I-AUC and glucose concentration from 14 to 24 h postnatal did not differ between treatments. However, we acknowledge that we are limited in our ability to estimate glucose disposal between treatments, as we did not measure intestinal glucose oxidation (Reynolds, 2002), hepatic glycogen deposition (Welboren et al., 2021b), or renal glucose clearance (van Meirhaeghe et al., 1988; Ferrannini et al., 2020). Because we do not have an estimate of glucose disposal respective to absorption or direct measurement of luminal to ad luminal glucose flux, it is uncertain if increased lactase enzymatic activity altered the total glucose available for absorption.

We hypothesized that colostral insulin would be absorbed across the GIT, such that there would be increases in plasma insulin and suppression of plasma glucose. Endogenous insulin secretion was not confounded in this study by unequal macronutrient intake. Despite these measures, we did not observe that plasma insulin was increased following the first postprandial period, nor did we observe that plasma glucose concentration decreased proportionally with increasing colostral insulin dose. Although we did observe moderate changes in insulin clearance during the first and second postprandial period and an increase in maximum plasma insulin concentration during the second postprandial period, these effects were negligible and more reasonably related to shifts in glucose rather than insulin absorption. In addition, we dosed a FITC-insulin (human) as a proportion (20%) of the total insulin supplemented in the 5BI and 10BI calves to distinguish our exogenous insulin sources from the endogenous secretion; however, our fluorescent label was not adequately sensitive to distinguish from endogenous fluorophores (for example, NADH/NADPH and flavins; Lakowicz, 2006) in plasma. As such, we were unable to use our fluorescent tag to estimate the rate of absorption of exogenous insulin across the intestinal tract. For future work, higher doses of FITC-insulin would be necessary to differentiate FITC-derived fluorescence from autofluorescence. The lack of increase in postprandial insulin concentrations and corresponding decreases in plasma glucose and serum NEFA in response to insulin supplementation suggests that insulin may not cross the neonatal bovine GIT into the periphery. In contrast to what has been observed for other neonatal mammalian species (Mosinger et al., 1959; Kelly, 1960, Asplund et al., 1962), it is possible that the neonatal

bovine GIT is impermeable to insulin. Yet, others have observed hypoglycemic and hyperinsulinemic responses when insulin is supplemented in milk (Pierce et al., 1964) or before enteral feeding (Kirovski et al., 2008). Kirovski et al. (2008) administered 30 U of insulin/kg BW with or without glucose 30 min before enteral feeding and decreased blood glucose while increasing plasma insulin in calves. In comparison, Pierce et al. (1964) orally dosed 2 to 4 U/kg BW insulin solely, observing decreased blood glucose similar to Kirovski et al. (2008). The same authors also observed that 500 U of insulin/kg BW in milk caused depressions in blood glucose (Pierce et al., 1964). Comparatively, in this study, 5BI and 10BI provided 1.7 and 3.3 U of insulin/kg BW in colostrum, respectively. Collectively, these studies indicate that pharmacological doses of insulin administered in colostrum or milk will induce hypoglycemia in calves, whereas supraphysiological doses that mimic those found in colostrum will only induce the same effect when provided orally before enteral feeding and not when fed in colostrum. As such, we find that the native range of colostrum insulin concentration (6 to 327 μg/L; Malven et al., 1987; Blum and Hammon, 2000; Zinicola and Bicalho, 2019) may not be high enough to elicit a hypoglycemic-hyperinsulinemic response in neonatal calves.

Within this study, we observed temporal patterns in plasma glucose, serum NEFA, and plasma insulin that provide information on the relative metabolic and physiological state of the neonatal calf. Plasma glucose and serum NEFA concentrations are relatively similar to what has been previously reported for neonatal calves (Knowles et al., 2000), but still offer insight into neonatal calf metabolism. As previously described, neonatal calves are challenged for glucose homeostasis as they have limited glycogen supplies (Okamoto et al., 1986; Girard, 1990), relatively underdeveloped endogenous glucose production (Steinhoff-Wagner et al., 2011b; Hammon et al., 2012), and a lesser capacity to digest lactose and absorb glucose (Girard, 1990; Girard et al., 1992; Steinhoff-Wagner et al., 2011a) at birth. Contrasting plasma glucose concentrations following the first and second postprandial periods gives evidence to maturation of intestinal capacity for lactose digestion and glucose absorption (Steinhoff-Wagner et al., 2011a), as the rate of plasma glucose increase is lesser, and concentrations are more sustained throughout the second postprandial period. Plasma glucose concentrations during the second postprandial period never equilibrate to baseline but do so after the first colostrum meal. Plasma glucose is lowest at birth before colostrum feeding for neonatal calves (Knowles et al., 2000) due to the immediate cessation of placental glucose supply and lack of lactose consumption. We observed that plasma glucose was cleared so rapidly during the first postprandial period that by 6 h postfeeding concentrations had equilibrated to baseline (~65 mg/dL). The concurrent transient rise in serum NEFA at the same time point and moderate recovery of plasma glucose concentrations at 8 and 10 h postfeeding suggests that neonatal calves in this study began utilizing their own energy reserves for endogenous gluconeogenesis (Vermorel et al., 1989; Steinhoff-Wagner et al., 2011b; Hammon et al., 2012). Neonatal calves have elevated metabolic rates (Okamoto et al., 1986; Vermorel et al., 1983, 1989) and the use of body reserves for endogenous glucose production demonstrates that a secondary colostrum feeding within 6 h of the first is likely necessary to prevent body reserve catabolism.

With respect to postprandial plasma insulin concentration following the first and second meal, concentrations are low, as is typical for newborn calves (review by Hammon et al., 2012). Peak insulin and time-dependent concentrations were not largely different between the first and second postprandial periods, irrespective of the magnitude of difference in corresponding plasma glucose concentrations. Because plasma glucose is the primary stimulation for pancreatic insulin release, the similarity in plasma insulin curves after the first and second meals reflects the immaturity of the newborn calf pancreas (Grütter and Blum, 1991). At 5 h postnatal, we observed a second transient rise of lesser magnitude than the first peak in plasma insulin at 2.5 h postnatal. Following the second meal, a secondary transient increase in plasma insulin was not observed. It is unclear why this transient rise occurred without an apparent stimulus because plasma glucose concentration was decreasing at this time.

CONCLUSIONS

We demonstrated that colostral insulin downregulates mRNA expression of the insulin signaling pathway but stimulates gastrointestinal development, conferring downstream benefits with respect to energy metabolism due to enhanced lactose digestion and glucose availability. The native colostral insulin concentration is not sufficiently high enough to induce a hyperinsulinemic-hypoglycemic state in the newborn calf. As such, high-insulin colostrum should be considered beneficial and preferable for neonatal calf development. Furthermore, the temporal patterns in plasma glucose and serum NEFA concentration that we observed during the first 14 h postnatal suggest that newborn calves may require a second meal of colostrum within 6 h of the first feeding to prevent catabolism of body energy reserves.

ACKNOWLEDGMENTS

The authors express appreciation of the aid provided by K. Acton, L. Buss, E. Croft, K. Cruickshank, A. Fischer-Tlustos, M. Lam, M. Líevre, A. Lopez, S. Mellors, J. Morrison, L. Rostoll Cangiano, D. Seymour, A. Welboren, M. Williams, and T. Yohe (University of Guelph, Guelph, ON, Canada) in sample collection and animal husbandry. Special thanks is given to Joe Loewith and Sons, Ltd. (Lynden, ON, Canada) and their wonderful staff for their support throughout the experiment. Gratitude is extended to the National Science and Engineering Research Council (Ottawa, ON, Canada) for funding this project and to the Saskatoon Colostrum Company Ltd. (Saskatoon, SK, Canada) for their laboratory contributions. The authors have not stated any conflicts of interest.

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